Vangapandu Harish Naidu

Data Scientist

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PROFESSIONAL SUMMARY

Data Scientist with 2.5+ years of experience designing and deploying production AI systems on AWS and GCP. Specialized in LLM applications, RAG architectures, agentic frameworks, hybrid ML systems, and scalable GenAI solutions. Delivered enterprise-grade systems across education, logistics, media, aviation, and e-commerce domains — driving automation, reducing operational latency, and replacing brittle rule-based systems with robust, learning-based approaches.

TECHNICAL SKILLS

Core AI/ML	Python, SQL, Machine Learning, Deep Learning, NLP, LLMs, RAG, Recommender Systems, Fine-Tuning, Prompt Engineering, Generative AI, Agentic AI, XGBoost
Frameworks	LangChain, FastAPI, Flask, Boto3, TensorFlow, PyTorch, Skyfield, CelesTrak
AWS	Bedrock, Lambda, S3, DynamoDB, EC2, Step Functions, API Gateway, Aurora PostgreSQL, SQS, SageMaker, Kendra, Chime, RDS
GCP	Vertex AI (Embeddings), Cloud Storage
Computer Vision	OpenPose, Detectron2, DensePose, YOLO, DeepFace, MediaPipe
Data & DevOps	PostgreSQL, RDS, Git, Model Deployment, Monitoring, Docker

PROFESSIONAL EXPERIENCE

Minfy Technologies — Data Scientist

08/2023 – Present | Hyderabad

1. Satellite Ground Station Recommendation Engine

XGBoost · AWS SageMaker · Lambda · S3 · RDS · Skyfield · CelesTrak

- Developed a **hybrid recommendation engine** combining deterministic rule-based feasibility checks with XGBoost ML predictions to recommend optimal ground stations for satellite communication passes.
- Implemented rule-based filtering to enforce hard operational constraints: minimum elevation angle, wavelength compatibility, visibility window, and ground station availability.
- Trained an **XGBoost prediction model** to estimate link feasibility and communication quality using features from orbital geometry, pass duration, elevation profiles, atmospheric conditions, and weather signals.
- Leveraged **Skyfield** for real-time satellite position and velocity tracking; used **CelesTrak** for automated TLE updates to maintain orbital accuracy.
- Designed a hybrid scoring mechanism combining rule validation with ML prediction scores to rank feasible ground stations per satellite pass window.
- Deployed trained model as a **SageMaker endpoint**; model artifacts stored in **S3**; endpoint invoked via **AWS Lambda** for serverless inference. Used **RDS** to persist ground station and satellite metadata.
- Built an end-to-end pipeline processing user inputs (satellite, date, time window) to deliver ranked ground station recommendations in near real-time via a web-based workflow.

2. AI-Based Question Paper Generation & Exam Proctoring System

Amazon Bedrock (Claude Sonnet 3.7) · Step Functions · Lambda · Aurora PostgreSQL · DynamoDB · SQS · EC2 · API Gateway · Twilio · YOLO · DeepFace · MediaPipe

- Designed and deployed an LLM-based assessment engine using **Claude Sonnet 3.7 on Amazon Bedrock** with parallel **Step Functions** orchestration, reducing 200-question paper generation from ~10 min to ~1 min (~90% latency improvement).
- Built a vector-search knowledge base (syllabus + past papers) on **Aurora PostgreSQL** with metadata indexing to improve retrieval speed and contextual relevance.
- Integrated a sandboxed Python execution environment to auto-generate diagrams and equations (Matplotlib) with <20s turnaround and S3-based delivery.
- Implemented full workflow via **AWS Lambda and API Gateway**; integrated **Twilio WhatsApp bot** for school users with **DynamoDB** maintaining persistent chat history for multi-turn conversations.
- Designed a **human handoff mechanism** — when the bot cannot resolve a query, it escalates the conversation to an admin dashboard for manual resolution without dropping context.
- Built an AI proctoring module deployed on **EC2** with **SQS for sequential request processing**; used **YOLO** to detect electronic gadgets, head rotation, looking-away behavior, and multiple persons in frame; used **DeepFace and MediaPipe** for identity verification and behavior analysis.

3. Swayam – No-Code Agentic Bot Platform

LangChain · FastAPI · Vertex AI · GPT-3.5 / GPT-4o · Gemini · AWS · GCP

- Architected a **no-code agentic platform** enabling organizations to build and deploy custom AI bots without coding, using LangChain-based orchestration with tool calls and RAG pipelines.
- Supported multiple LLMs — **GPT-3.5, GPT-4o, and Gemini** — with flexible model selection per use case and client requirement.
- Built helpdesk and interview bots with adaptive multilingual support across text and voice channels for enterprise environments.
- Implemented custom RAG retrieval supporting **documents, web links, and code repositories**; tuned embedding similarity thresholds to improve response accuracy and reduce hallucinations.
- Used **Vertex AI embeddings** on GCP for high-quality custom vector representations; deployed backend services via **FastAPI** across AWS and GCP infrastructure.
- Platform successfully adopted by enterprise clients: **DTDC, Sun TV, and IndiGo Airlines**.

4. Gromme.ai – Virtual Try-On Fashion E-Commerce Platform

AWS SQS · DynamoDB · S3 · EC2 (G5/C5) · Amazon Kendra · Amazon Chime · OpenPose · Detectron2 · DensePose

- Developed a full-stack e-commerce platform with an end-to-end virtual try-on pipeline for mobile users, enabling high-quality garment overlay across diverse body types.
- Built a custom image preprocessing script that standardizes garment and body images without stretching or compression, significantly improving alignment accuracy and overall try-on quality.
- Engineered a precise **body masking pipeline** for user images, accurately isolating body regions to enable seamless and realistic garment reflection and overlay.
- Built an image processing pipeline using **OpenPose, Detectron2, and DensePose** for pose detection and body segmentation; all preprocessed images stored and managed in **S3**.
- Implemented asynchronous processing using **AWS SQS and DynamoDB** for request tracking and scalability; optimized inference on **AWS G5 and C5 instances** achieving ~40s per image processing time.
- Implemented a **hybrid search bar** functioning simultaneously as a product search engine and conversational recommendation bot, powered by **Amazon Kendra** for garment metadata indexing and retrieval.
- Integrated **Amazon Chime** for in-platform group chat, enabling users to invite friends, share try-on results, and collaborate; **DynamoDB** used for persistent chat history.
- Conducted human validation testing with ~1,000 garment images and shared results with the customer for quality sign-off prior to production launch.

CERTIFICATIONS

✓ AWS Certified Machine Learning – Specialty (MLS-C01)

✓ AWS Certified Machine Learning Engineer (MLA-C01)

✓ AWS Certified AI Practitioner (AIF-C01)

✓ Google Professional Machine Learning Engineer (Associate)

EDUCATION

Bachelor of Technology (B.Tech) 2018 – 2022 | Visakhapatnam

Anil Neerukonda Institute of Technology and Sciences